

**Building sector flow chart for St. Paul
(Residential, commercial, and industrial)**

Inputs for energy use calculation

- Energy use intensity by floor area in homes
- Number of single- (SFM) and multi-family (MFM) homes
- Avg floor area: SFM ~1800 sf and MFM ~1400 sf
 - Trend to 2040 will be city specific
- Number of workers in industrial and commercial sectors
- Energy use per workers in commercial and industrial sectors



Equation for energy use with term defined

$Energy_{res} = EUI_{res} * (FA_{SF} + FA_{MF})$: $Energy_{res}$ is total annual energy use in homes. EUI_{res} is energy use intensity by floor area. FA_{SF} is total floor area of single-family homes; FA_{MF} is total floor area of multi-family homes.

$Energy_{com} = EUI_{com} * Wk_{com}$: $Energy_{com}$ is total energy use in commercial sector (by energy types). EUI_{com} is energy use per worker in commercial sector. Wk_{com} is the number of workers in commercial sector.

$Energy_{ind} = EUI_{ind} * Wk_{ind}$: $Energy_{ind}$ is total energy use in industrial sector (by energy types). EUI_{ind} is energy use per worker in industrial sector. Wk_{ind} is the number of workers in industrial sector.

Inputs for GHG calcs w assumptions

- Energy end-use by types (natural gas or electricity) in homes/businesses/industry
- GHG emission factor (EF) for energy carrier. For electricity grid, 80% reduction in EF compared to 2018 grid (from ~1500 lb/MWh to 250 lb CO₂/MWh). For gas, 25% of natural gas is from renewable.



Equation for GHG calculation

$$GHG = Energy * EF$$

Energy is total end-use energy in different sector by energy use type (electricity or natural gas)

EF is emission factor. Natural gas' EF is emissions from the combustion. Electricity's EF is emissions at the grid level, extracted from eGrid data.

Inputs for cost and energy saved

See cost table.

Fossil fuel cost trend to 2040; cost of construction (SFM and MFM) efficiency improvement per sf.

Levelized cost of zero-carbon district energy, zero-carbon electric grid, renewable natural gas (NG).



Equation for cost calculation

Cost = Cost per unit of appliance/improvement * total number of appliance (or total area of improvement)

Savings = total unit of energy use saved * cost of per unit energy



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List of interventions	Two-line description in simple language	How calculated
A. Increase multi-family homes to 50% share of new construction (in compact development scenario relative to BTP).	More compact land use can result in more multi-family homes. Based on our analysis in the region, multi-family home is smaller in size than single-family homes. Furthermore, a neighborhood with more multi-family floor area has lower energy use intensity.	First, calculate the floor area change due to this policy. Second, calculate the impact of change in the percentage of multi-family home floor area on energy use intensity by applying an elasticity factor. Third, calculate household energy use by multiplying the new floor area with the new EUI. Forth, calculate the difference from the baseline trend projection.
B. Compute energy impact of increase tree canopy to various levels (choose from: doubling tree canopy area; additional 1 tree per 4-person target per city; convert all pervious to trees; convert 80% impervious roads and parking lots to tree cover).	Tree canopy can sequester carbon and reduce energy use. This module only calculates energy use using elasticity factor from analysis in St. Paul's data (1% increase in tree canopy will reduce ~0.7% electricity use intensity).	First, calculate the impact of change in tree canopy coverage on energy use intensity by applying an elasticity factor. Second, calculate energy savings by multiplying the difference in EUI and total residential floor area.
C. Double % of homes to energy affordability recognizing electricity cost will be more than doubled in a net-zero grid in 2040.	Cities can control new houses' floor area based on the energy affordability criterion that 6% of household income is spent on energy bills (ACEEE, 2020).	First, calculate the floor area cap based on energy poverty definition using current EUI and median household income. Second, calculate floor area change due to this policy. Third, calculate energy savings by multiplying EUI after intervention B with floor are change.
D. Increase new buildings to be high energy efficient (e.g., LEED-Gold).	Implement strict building codes for new homes, requiring energy use intensity to be 25 kBtu/sqft.	Multiply the difference in EUI with the floor areas of new buildings that are highly energy efficient.
E. Existing buildings become energy efficient	Retrofit existing buildings to reduce energy use intensity by 33%.	Multiply the difference in EUI with the floor areas of existing buildings that will be retrofitted.
F. Largest existing buildings and commercial buildings meet highest efficiency standards	Deep retrofit existing buildings targeting those with the largest size to reduce energy use intensity by 66%.	
G. Behavioral nudging	Cities can have mandates to inserting effective messages on bills and promoting the adoption of real-time energy use displays and smart meters to reduce energy use.	Calculate energy use saving by multiplying total household energy use with the percentage of homes influenced and the effectiveness of these energy nudges in a sequential way. Messaging is the first, in-home real-time energy displace is the second, and installing smart meters is the last.

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H. Zero-carbon district energy for higher density areas	Existing district energy systems	Multiply current carbon emissions with the percentage of reduction as planned.
I. Expanding district energy systems	With the development of technology, advance district energy system can utilize various low-/zero-carbon energy sources, e.g., heat of sewage, and exhausted heat from industrial process, to be zero-carbon.	First, calculate the amount of waste heat that is available from sewage and industrial processes. Second, assume that these waste heat can be all utilized in district energy system and calculate the amount of natural gas for commercial heating that can be replaced.
J. Decarbonize grid	Electric grid will be decarbonized as proposed by the electric utility in Minnesota.	Apply the change of grid-level carbon emission factor to electricity use after interventions A to I.
K. Electrifying heating	Space heating and water heating can be electrified via efficient heat pumps and electric water heaters or heat pump water heaters.	First, calculate the gas heating demand. Second, calculate the amount of new electricity heating demand due to electrification. Last, calculate the amount of gas reduced.
L. Using renewable natural gas	Gas utilities can use renewable natural gas from various waste streams, e.g., food waste, to replace natural gas use.	Apply Minnesota state-level renewable natural gas potential to the rest of gas use after electrifying heating to calculate the amount of carbon reduced when utilizing renewable natural gas.

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